

Emotionally intelligent preference optimization for conversational behavior change support

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Abstract

Procrastination negatively affects productivity and well-being, yet current large language models (LLMs) often lack the nuanced emotional understanding necessary for effective behavior-change support. This study proposes emotionally intelligent preference optimization (EIPO), a unified framework integrating emotional chain-of-thought reasoning, teacher-guided self-correction, and direct preference optimization to enhance emotional intelligence in conversational user interfaces. Through ablation experiments evaluated using the emotional generation score (EGS), we demonstrate that EIPO significantly improves empathetic and constructive responses in procrastination-related interactions. Crucially, cross-model validation on Qwen2.5-7B, Llama-3.2-3B, and Gemma-2-2B confirms EIPO's model-agnostic design, with consistent EGS gains ($\Delta = +0.13$ to $+0.27$) and robust independent evaluation using Claude-3.5-Sonnet. These findings establish EIPO as a principled, transferable framework for embedding emotionally intelligent behavior-change support into diverse LLM-driven CUIs.

Introduction

Current large language models (LLMs) often lack nuanced emotional understanding, which can undermine the effectiveness of conversational interfaces in supporting users' behavior change, such as managing procrastination^{1,2}. This study draws on prior work in sentiment analysis, computational emotion detection, self-correcting and preference-optimized LLMs, emotion-aware language models, and evaluation frameworks for emotional quality in conversational AI. These foundations inform how emotional intelligence can be modelled, enhanced, and assessed in behavior-change conversational AI. This paper contributes by investigating how deeper emotional reasoning, self-correction, and preference learning can strengthen an LLM's ability to support users struggling with procrastination.

Existing conversational AI systems for behavior change often rely on simple sentiment cues or motivational prompts, offering only partial emotional intelligence^{3–8}. It remains unclear how different approaches to emotional reasoning, such as structured chain-of-thought prompting versus classifier-based methods, impact the effectiveness of AI support for sustainable behavior change, including procrastination mitigation and energy consumption reduction^{9–16}. The impact of self-correction techniques on emotional reasoning in such contexts is also underexplored¹⁷. Likewise, it is not yet known whether preference-based fine-tuning methods meaningfully improve the

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emotional intelligence of LLMs in procrastination-related interactions. These open questions suggest a need for systematic investigation into how reasoning, self-correction, and preference learning might contribute to more emotionally supportive conversational AI, particularly in contexts that require nuanced, user-centered guidance.

To address these gaps, we propose emotionally intelligent preference optimization (EIPO), a unified framework that integrates structured emotional reasoning, self-correction, and preference-based fine-tuning. EIPO operates in three stages: (1) an emotional reasoning module uses emotional chain-of-thought (ECoT) to generate context-aware, empathetic responses; (2) a self-correction module employs a teacher model to refine these responses, producing high-quality preference pairs; and (3) a preference optimization module fine-tunes the student model via direct preference optimization (DPO) on these pairs, embedding emotional intelligence directly into the model's parameters. We evaluate EIPO through a series of comparative analyses that isolate the contribution of each component, using the emotional generation score (EGS) and human-validated win-rate metrics. Our results demonstrated that while emotional reasoning alone outperforms classifier-based methods, self-correction raises the ceiling of response quality, and DPO stabilizes these gains, yielding a final model that is both more emotionally intelligent and more consistent. Besides, a key advantage of the proposed emotionally intelligent preference optimization (EIPO) framework is its model-agnostic design. Rather than relying on a specific LLM architecture, EIPO integrates emotional chain-of-thought reasoning, teacher-guided self-correction, and direct preference optimization as modular components that can be applied to a broad range of instruction-following language models. Empirically, we validate this claim by applying EIPO to two additional models, Llama-3.2-3B-Instruct and Gemma-2-2B-IT, alongside Qwen2.5-7B-Instruct. All three models exhibited consistent EGS improvements ($\Delta = +0.13$, $+0.26$, and $+0.27$, respectively), confirming that EIPO's staged optimization generalizes across architectures.

Taken together, the results demonstrate that emotional intelligence in conversational AI is most effectively enhanced through the synergistic integration of emotional reasoning, reflective self-correction, and preference optimization, rather than through any individual technique alone. Emotional chain-of-thought provides explicit emotional reasoning to improve contextual understanding, self-correction transforms this reasoning into high-quality preference supervision through iterative refinement, and preference optimization internalizes these learned preferences into the model parameters, enabling more consistent emotionally intelligent responses during inference. This progressive optimization process systematically transfers emotional reasoning from prompt-level guidance to intrinsic model behavior. By embedding these capabilities into conversational user interfaces (CUIs), the proposed EIPO framework substantially improves an LLM's ability to provide empathetic and constructive support for users experiencing procrastination while offering a principled methodology for developing emotionally intelligent conversational agents for behavior-change applications.

Our contributions in this paper are:

- Introduces a principled framework (EIPO) for embedding emotional intelligence directly into Conversational User Interfaces through structured reasoning, self-correction, and preference optimization.
- Develop and release a preference-aligned dataset for emotional intelligence, constructed from self-corrected emotional responses tailored to procrastination-related interactions.
- We demonstrate that the staged optimization (reasoning $\rightarrow$ refinement $\rightarrow$ preference optimization) produces cumulative improvements in emotional support quality through component-wise ablation experiments.

Related work

Foundations of emotion detection in LLMs

Integrating emotion detection into chatbots can enhance user experience by recognizing human emotions such as anger, happiness, or sadness. Emotions are expressed through text and visual elements, including emojis, word

choice, formatting, and color^{18–21}. Linguistic cues can reveal personality, motivation, self-efficacy, and barriers to sustainable behaviors, enabling AI to tailor interventions^{3,4}. Visual cues may complement or contradict text, so careful interpretation of multimodal signals is crucial²². Sentiment analysis at sentence, document, or aspect levels extracts attitudes from text, capturing both general affect and behavior-specific sentiments^{8,23–25}. Emotion classification categorizes emotions via categorical or dimensional models, such as valence, arousal, and dominance^{6–8}.

Emotion detection methods are typically divided into lexicon-based and machine learning (ML) approaches. Lexicon-based methods rely on sentiment lexicons, bag-of-words models, or statistical techniques like latent semantic analysis (LSA) and the hyperspace analogue to language (HAL) to identify emotions in text^{9–12}. ML-based approaches use supervised or unsupervised algorithms: supervised methods learn from labeled datasets, including manually annotated corpora or automatically labeled tweets via hashtags and emoticons^{13–16}, while unsupervised methods detect latent structures in unlabeled data for emotion classification.

Nowadays, emotion detection is increasingly integrated into LLMs. Some studies focus on text-based emotion detection, while others deal with visual emotion analysis⁵. Emotion detection enhances AI systems' ability to recognize and respond to users' feelings, aiding in the diagnosis of mental health conditions²⁶. Key applications include emotion-aware wearable devices and mental health chatbots²⁷. Another area of research is to embed emotion into LLMs and give them specific persona to make responses more emotionally intelligent⁵. Modular emotional intelligence enhancement (MoEI) is also proposed as a method to enhance a LLM's emotional intelligence while preserving its general intelligence²⁸. It uses low-rank adaptation (LoRA) to add extra trainable modules for tasks related to emotional intelligence. It outperforms naive fine-tuning and standard LoRA on emotional intelligence, and preserves general intelligence than both²⁸.

Previously, prompts relied heavily on specific patterns tailored to particular machine learning models. However, with the advancement of large-scale language models, prompting has become more flexible and accessible⁵. These models allow us to input prompts in various styles and tones, mimicking natural human speech. Emotions are always conveyed in human speech. Some examples of prompts are demonstrated below:

1. Classify the following text into one of the six basic emotions: happiness, sadness, anger, fear, surprise, disgust.
2. Detect all emotions expressed in the text. If multiple emotions are present, list them all.
3. Identify the user's emotion in the following sentence and explain how it might influence their motivation for sustainable behavior change.

Through writing prompts for LLMs, people can express their emotions consciously and unconsciously. These emotions can be conveyed in multiple ways, ranging from subtle hints to more explicit expressions²⁹. Zhong et al.³⁰ showed that ChatGPT performs at a level comparable to fine-tuned BERT models. However, Zhang et al. conducted a more extensive evaluation across multiple sentiment analysis tasks. Their findings indicate that although models perform poorly in fine-grained sentiment analysis, they demonstrate strong zero-shot abilities in simpler tasks like binary sentiment classification.

Self-correction of LLMs for emotion detection

Self-correction enables LLMs to refine their outputs during inference, improving reasoning, style, and overall response quality^{31–34}. It generally involves three stages: initial response generation, feedback generation, and response refinement^{35,36}. Beyond explicit feedback loops, alternative methods include direct refinement, which enhances responses without explicit feedback³⁷, and post-hoc correction, which revises outputs after generation³⁸. Generation-time correction (step-level correction) improves reasoning by providing feedback during intermediate steps^{39,40}, while cross-model correction employs different models to evaluate or refine responses, often in multi-agent or fine-tuned model setups⁴¹.

In terms of sources of feedback, intrinsic self-correction utilizes prompt engineering to ask LLMs to generate feedback themselves. Prompting strategies encompass methods such as basic zero-shot or few-shot prompts,

breaking down responses into components, and assessing confidence levels⁴¹. Self-correction can also be connected to external tools such as code executors and external knowledge from search engines⁴¹. Also, model fine-tuning can be served as a feedback for self-correction. Models are trained through supervised fine-tuning or reinforcement learning⁴¹.

However, when it is related to reasoning tasks for LLMs, they may often have difficulty in recognizing and correcting their own mistakes without external feedback, as mentioned in³⁶. Additionally, efforts to self-correct logical or reasoning errors can occasionally alter correct responses into incorrect ones, leading to a decline in overall performance³⁶. Yang et al., proposed a new cross-model paradigm to enhance the capability of LLMs for both reasoning and self-correction by leveraging external model feedback¹⁷. Thus, it overcomes the limitations of LLMs' initial responses and expands their ability to tackle a broader spectrum of problems. This method has not yet been applied to emotion detection self-correction in LLMs, highlighting a potential research gap worth exploring.

Direct preference optimization (DPO) for emotion detection

Reinforcement learning from human feedback (RLHF) plays a crucial role in training today's most advanced large language models. The algorithm adjusts the language model's responses to better reflect human preferences^{42,43}. RLHF involves collecting human preferences to train a reward model, then using reinforcement learning to adjust the language model to produce preferred responses while staying close to its original behavior⁴⁴. However, RLHF has drawbacks for two reasons. Firstly, the reward model is typically as large as the language model, making it costly to train and maintain⁴⁴. Secondly, reinforcement learning algorithms are highly sensitive to hyperparameter settings, making them difficult to fine-tune⁴⁵.

A simple effective alternative to RLHF is direct preference optimization (DPO). DPO is a method for training language models to align with human preferences directly, without relying on explicit reward modeling or reinforcement learning⁴⁶. It is an approach that implicitly targets the same objective as existing RLHF algorithms, maximizing rewards with a KL divergence constraint, while being simple to implement and easy to train⁴⁶. Whereas traditional methods rely on a preference model to compute a preference loss for training a reward model, followed by optimizing a policy based on that reward model, DPO reformulates the process by expressing the preference loss directly in terms of the policy. Therefore, DPO turns the human preference signal into a direct learning signal for the policy, bypassing the separate reward modeling step used in traditional RLHF. Leveraging the dataset of human preferences over model responses, DPO can optimize the policy with a straightforward binary cross-entropy objective, without the need to explicitly learn a reward function or sample from the policy during the training⁴⁶.

However, the emotional intelligence of LLMs in the context of DPO remains relatively underexplored, with limited research examining how effectively LLMs can be aligned to exhibit emotionally intelligent behaviors through preference-based fine-tuning, particularly in tasks involving procrastination avoidance. Moreover, the self-correction capabilities of the teacher model can be leveraged to construct the dataset for DPO training by generating refined response that better align with human-preferred qualities, thereby serving as high-quality references for preference-based learning.

Iterative feedback, self-correction, and preference optimization

Recent studies have demonstrated the effectiveness of iterative feedback and preference-based optimization for improving the capabilities of large language models across different application domains. For example, See2Refine proposes a human-free closed loop refinement framework in which a vision-language model (VLM) provides iterative feedback to improve LLM-generated external human-machine interface (eHMI) actions, showing that automatic feedback can effectively enhance generation quality without requiring human annotation⁴⁷. Similarly, active learning for abstractive text summarization via LLM-determined curriculum and certainty gain

maximization introduces an active learning framework that leverages LLM-generated curricula and certainty gain estimation to select informative training samples, thereby improving fine-tuning efficiency while reducing annotation costs⁴⁸. More recently, TimeClaw presents a time-series AI agent that progressively improves its reasoning ability through exploratory execution learning, demonstrating the benefits of staged optimization and iterative experience refinement⁴⁹. These studies highlight the growing trend of integrating automated feedback, iterative refinement, and preference-based learning to enhance LLM performance. In contrast, the proposed emotionally intelligent preference optimization (EIPO) framework applies these principles specifically to emotionally intelligent conversational behavior-change support by combining emotional chain-of-thought reasoning, teacher-guided self-correction, automatic emotional preference-pair construction, and Direct Preference Optimization to progressively transform emotional reasoning into trainable preference supervision.

Evaluation frameworks for emotion detection in conversational AI

Emotion detection in conversational AI systems can be evaluated through multiple frameworks, including machine learning metrics, user experience assessments and real-world application performance. Machine learning metrics is also known as model assessment⁸. Another approach to assess the performance of emotionally intelligent conversational AI is by using human evaluators⁵⁰. While automatic evaluation is more efficient and involves less overhead than human evaluation, it does not take into account whether the generated emotional response is suitable and natural. Human evaluation is typically assessed using a Likert scale. Several studies employed Fleiss' kappa test to assess annotator agreement and consistency in ratings.

The Emotional generation score (EGS) is proposed as a novel method by Li et al.⁵¹, for evaluating the emotional output of LLMs, applicable to both automatic and manual evaluation, and the research found that EGS can function as a trustworthy automated evaluation method, making expert annotation unnecessary. It is an answer-free automatic method that integrates Goleman's emotional intelligence theory⁵² as a representation of human expert consensus. EGS assesses the emotional generation capability of LLMs from various aspects of human emotional intelligence. The criteria for EGS are as following:

- C1: The speaker's reply should align with the given context.
- C2: The speaker's response should avoid conveying negative emotions toward anyone or anything, including disgust, resentment, discrimination, or hatred.
- C3: The speaker's response should convey positive emotions, such as happiness, empathy, support, advice, encouragement, and comfort.
- C4: The speaker's response should contribute positively to the conversation's progress, such as easing negative emotions for both parties and strengthening their relationship.

Based on the literature review, we will adopt emotional chain-of-thought (ECoT), self-correction, and direct preference optimization (DPO) to enhance the emotional intelligence of LLMs for procrastination mitigation. The Emotional Generation Score (EGS) will be employed as the primary evaluation metric to assess the effectiveness of the proposed approach.

The EIPO framework and research hypothesis

Based on the literature review, we propose emotionally intelligent preference optimization (EIPO), a unified framework that enhances LLM emotional intelligence through three integrated stages: (1) structured emotional reasoning via emotional chain-of-thought (ECoT), (2) self-correction for response refinement, and (3) preference-based fine-tuning via DPO. Rather than treating these as isolated interventions, we conceptualize EIPO as a single pipeline in which each successive stage builds upon the last.

While prior work has examined emotional reasoning, self-correction, and preference optimization independently, no study has investigated their sequential integration for behavior change support. We address this gap through a component-wise ablation design that isolates the marginal contribution of each stage.

Hence, the hypotheses can be proposed as:

- *H1 (Emotional reasoning hypothesis)* Within the EIPO framework, the emotional chain-of-thought (ECoT) stage generates responses with significantly higher perceived emotional intelligence, as measured by the emotional generation score (EGS), than an equivalent framework that replaces ECoT with classifier-based emotion prompting.
- *H2 (Self-correction hypothesis)* Augmenting the EIPO framework with a self-correction stage, wherein a teacher model refines the student's ECoT-generated responses, yields significant improvements in EGS scores compared to the framework with ECoT alone.
- *H3 (Preference optimization hypothesis)* Fine-tuning the student model via direct preference optimization (DPO) on preference pairs constructed from the self-correction stage produces significant EGS gains and a positive pairwise win-rate against the base ECoT model.

Dataset

EmpatheticDialogues is a conversation-based dataset that can be leveraged in studies on emotional response⁵³. The dataset consists of 24,850 conversations centered around situation descriptions, collected from 810 unique participants⁵³. The dataset was further refined to include only dialogues related to procrastination. The filtered procrastination-related dataset consists of 273 dialogues and examples are demonstrated below:

Example 1 (Conversation contexts):

User: "Yesterday was a great lazy day. I did nothing but sit outside in the sun and read a book. It was completely happy." **System:** "That sounds like a wonderful time. I wish I had the time for that!" User: "It is always a good idea to make time for some me time. It helps relax and clear your head."

Example 2 (Conversation contexts):

User: "My kids are so bad about bringing their dirty dishes to the sink and not rinsing them off." System: "Aw man, kids can be SO lazy and don't like to do what they're supposed to. Have you tried a rewards system to get them to do what they have to, or are they too old for that?" User: "They are too old for that. I don't have a dishwasher and have told them if they don't get it together they will be doing the dishes themselves."

The dataset contains 273 dialogues spanning 31 distinct emotional categories, providing broad emotional coverage. The most frequently represented emotions include annoyed (30 dialogues, 11.0%), guilty (22, 8.1%), angry (18, 6.6%), ashamed (16, 5.9%), content (13, 4.8%), apprehensive (13, 4.8%), anxious (11, 4.0%), and disappointed (11, 4.0%). Less frequent emotions, including hopeful, excited, nostalgic, and confident, were intentionally retained to increase emotional diversity and expose the model to less common affective states.

The situations represented in the dataset cover a wide range of real-world contexts that influence motivation and task completion, including academic work, workplace responsibilities, interpersonal relationships, health concerns, family obligations, unexpected life events, and emotional distress. This diversity was intended to expose the model to multiple forms of motivational barriers rather than limiting training to a single procrastination scenario.

Although the dataset is smaller than large-scale instruction tuning corpora, collecting high-quality procrastination-related conversational data presents significant practical challenges. Unlike general dialogue datasets,

behavior-change conversations often involve sensitive personal experiences, emotional struggles, and motivational difficulties. Data collection therefore requires careful ethical consideration, participant consent, and extensive quality control to ensure both authenticity and privacy.

Experimental evaluation of EIPO

Component 1: emotional reasoning

The first stage of the EIPO framework is emotional reasoning: the process by which the model interprets a user's emotional state and strategically structures its response to be empathetic, contextually appropriate, and behaviorally supportive. While prior work has primarily pursued emotional reasoning through external classifiers⁵⁴, our framework implements structured reasoning via emotional chain-of-thought (ECoT). We describe both approaches below, as the classifier-conditioned method serves as our baseline for evaluating the contribution of ECoT.

Incorporating a pre-trained emotion detection model into the prompt of an LLM is a commonly adopted approach to enhance the emotional intelligence of conversational AI systems⁵⁴. This strategy allows the LLM to dynamically adjust its responses based on the detected emotional state of the user, thereby improving the relevance, empathy, and appropriateness of its replies. By explicitly recognizing emotions such as frustration, anxiety, or motivation, the chatbot can better tailor its tone, language, and content to the user's needs, fostering a more human-like and supportive interaction⁵⁴. The pre-trained model used in this experiment is the fine-tuned DistilRoBERTa-base for emotion classification⁵⁵. DistilRoBERTa-base is a transformer model that was trained for the task of sentiment analysis. The model classifies inputs into seven categories, comprising six Ekman emotions—anger, disgust, fear, joy, sadness, and surprise—along with a neutral category⁵⁵. DistilRoBERTa was selected as the emotion classifier because it offers a favorable balance between classification accuracy, computational efficiency, and inference speed, making it suitable for conversational user interfaces that require real-time emotion recognition⁵⁵. Compared with larger transformer models, DistilRoBERTa substantially reduces computational cost while maintaining competitive performance on emotion classification tasks.

Emotional chain-of-thought (ECoT) was proposed by Li et al.⁵¹. It adopts a sequence of emotion-informed reasoning steps because this structure makes the model explicitly analyze user emotions and response strategies. These particular steps are chosen over others as they directly support emotionally intelligent dialogue generation. LLMs can perform the following steps for thinking⁵¹:

Step 1: [Understanding context] Describe what the user is experiencing or communicating based on their message.

Step 2: [Recognizing others' emotions] Identify what emotions the user might be feeling. Support your answer with specific clues from the message.

Step 3: [Recognizing self-emotions] Reflect on what tone and emotional stance you, as the assistant, should adopt to best support the user.

Step 4: [Managing self-emotions] Think about how to regulate your emotional expression to remain calm, constructive, and non-judgmental.

Step 5: [Influencing others' emotions] Plan how your response can improve the user's emotional state or move the conversation in a more positive direction.

This approach explicitly decomposes the complex process of emotional response into a step-by-step sequence, which not only unveils the LLMs' underlying "thought process," but also enables the model to break down the intricate task into manageable sub-tasks, thereby enhancing its performance in emotional generation. ECoT uses psychological theory for deeper emotional reasoning and alignment, and is specialized for emotion via Goleman's theory and expert guidelines, compared with traditional emotional generation models and standard

chain-of-thought. The experimental results indicate that ECoT substantially enhances LLMs' abilities in both understanding and generating emotionally intelligent responses⁵¹.

In this experiment, EGS will serve as the primary evaluation metric for assessing the emotional intelligence of conversational AI responses. We choose EGS because it can quantify the improvement that the ECoT method provided and compare different LLMs on emotional generation tasks fairly. Besides, it can also automate the evaluation process to handle large-scale experiments and ground evaluation in established psychological theory to ensure validity⁵¹. GPT-4 Turbo will be employed as the automated evaluator to ensure consistent and scalable assessment. GPT-4 Turbo is chosen for its superior consistency, interpretive accuracy, and efficiency in large-batch language assessment tasks. All hyperparameters, decoding settings, and other configuration details were kept at their default values unless otherwise specified. The evaluation will be guided by a carefully designed prompt tailored for GPT-4 Turbo, which outlines the EGS criteria and scoring instructions in detail. And the prompt examples are demonstrated below. We will additionally conduct human annotation to report the Cohen's kappa agreement score.

Prompt Example for emotional chain-of-thought

You are an emotionally intelligent assistant designed to support users in sensitive and emotionally challenging situations, such as procrastination, stress, self-doubt, or interpersonal conflicts.

Before generating your response, follow these reasoning steps to simulate high emotional intelligence, inspired by Goleman's framework

Step 1: [Understanding context] – Interpret what the user is experiencing or trying to communicate. What is the situation, concern, or emotional need?

Step 2: [Recognizing user's emotions] – Identify the likely emotions the user is feeling. Provide reasoning based on clues from their message (e.g., word choice, tone, or context).

Step 3: [Setting your emotional stance] – Determine the most appropriate tone, mindset, and emotional posture you should adopt in response (e.g., calm, encouraging, empathetic).

Step 4: [Managing your emotional expression] – Reflect on how to express your response constructively: avoid judgment, reinforce support, and remain emotionally balanced.

Step 5: [Positively influencing the user] – Decide how to move the conversation forward in a helpful, emotionally uplifting, and goal-directed way. What will help reduce distress, increase motivation, or build connection?

Final Step: [Emotionally intelligent response] – Generate a concise, warm, and constructive response to the user's message, applying all of the above reasoning.

Show all your reasoning steps first, then write the final emotional response.

Prompt Example for EGS evaluation

You are an expert evaluating the emotional intelligence of a chatbot's response using the emotional generation score (EGS) framework to evaluate the assistant response above. There are four evaluation criteria

C1: Context relevance – Does the response align appropriately with the user's message and situation?

C2: Emotional safety – Does the response avoid conveying any negative emotions toward anyone or anything, such as disgust, resentment, discrimination, or hatred?

C3: Positive emotion – Does the response convey positive emotions such as happiness, empathy, support, advice, encouragement, or comfort?

C4: Constructive contribution – Does the response help move the conversation forward in a positive way, such as by easing negative emotions or strengthening the relationship between the speaker and the user?

Please rate each criterion from 0 to 10, where

0 = completely fails the criterion

10 = fully satisfies the criterion

After scoring, provide a short explanation for each score.

Your evaluation

C1 (Context relevance): [0–10] – [brief explanation]

C2 (Emotional safety): [0–10] – [brief explanation]

C3 (Positive emotion): [0–10] – [brief explanation]

C4 (Constructive contribution): [0–10] – [brief explanation]

For the emotion classifier method, we adopt a two-stage pipeline. First, a fine-tuned DistilRoBERTa-base model predicts the emotion label for each input. This predicted label is then inserted into the prompt provided to GPT-3.5 Turbo. In this setup, GPT-3.5 Turbo does not perform emotion classification itself; instead, it conditions its generation on the externally supplied emotion category inferred by the classifier. GPT-3.5 Turbo was selected because it has been one of the most widely adopted and extensively evaluated large language models for conversational AI. Despite the emergence of newer models, GPT-3.5 Turbo remains a common baseline in dialogue

system research due to its strong conversational capabilities, cost-effectiveness, and broad accessibility through the OpenAI API.

In contrast, for the ECoT method, the prompt was provided directly to GPT-3.5 Turbo without an external classifier. For each emotional intelligence method applied to conversational AI, the dialogues are presented to the AI as real user input. The system then generates corresponding responses, which are subsequently evaluated by GPT-4 Turbo based on EGS prompt. Responses and evaluation results from this entire process are stored in a JSON file for subsequent statistical analysis. We manually annotated a randomly sampled 50% subset of the model outputs using the EGS metrics and computed Cohen's kappa to assess agreement between human annotations and the model's evaluation results. For the Emotion Classifier Model, the quadratic weighted kappa of 0.7478 indicates substantial agreement between human and model ratings. Similarly, for the ECoT method, the kappa score of 0.7649 also reflects substantial agreement, demonstrating strong alignment between human judgments and model-generated evaluations.

Component 2: self-correction

The second stage of the EIPO framework is self-correction. Based on the literature review of self-correction, we further enhanced our framework leveraging LLMs to enhance emotional intelligence using self-correction for procrastination avoidance. After the student model (GPT-3.5 Turbo) generates a response, the teacher model (GPT-4 Turbo) will provide an improved emotional response and feedback on the student model's output. We choose GPT-4 Turbo as teacher model because it has stronger reasoning, understanding, and generalization abilities. The initial student prompt can be referred to as prompt from Experiment 1. And the prompt for teacher model is:

Prompt for teacher model

You are a teaching model designed to evaluate and improve emotionally intelligent responses generated by a student assistant. The student assistant follows an emotional reasoning framework based on Goleman's emotional intelligence principles to respond to user messages about emotionally challenging situations, such as procrastination, stress, or self-doubt.

Your task

1. Carefully review the student assistant's reasoning steps and final response.
2. Identify weaknesses in emotion recognition, tone, supportiveness, or motivation
3. Generate a revised version of the final response that better demonstrates emotional intelligence.
4. Provide a brief explanation (feedback) about how and why your revision improves on the original.

Your output format:

[Improved emotional response]

< your revised emotionally intelligent response here >

[Feedback]

<explain what was missing, inappropriate, or suboptimal in the student's reasoning or final response, and how your version improves emotional intelligence and constructive contribution>

After receiving feedback from the teacher model, the student model is prompted to generate a revised response based on the teacher's suggestions and improvements. And the prompt is following:

Prompt for student model

You are a chatbot learning to improve emotional intelligence, especially for supporting users dealing with procrastination, stress, or self-doubt.

Your task: Based on the teacher's revision and feedback, generate a new, emotionally intelligent response in your own words. Your response should

Acknowledge and validate the user's emotional state

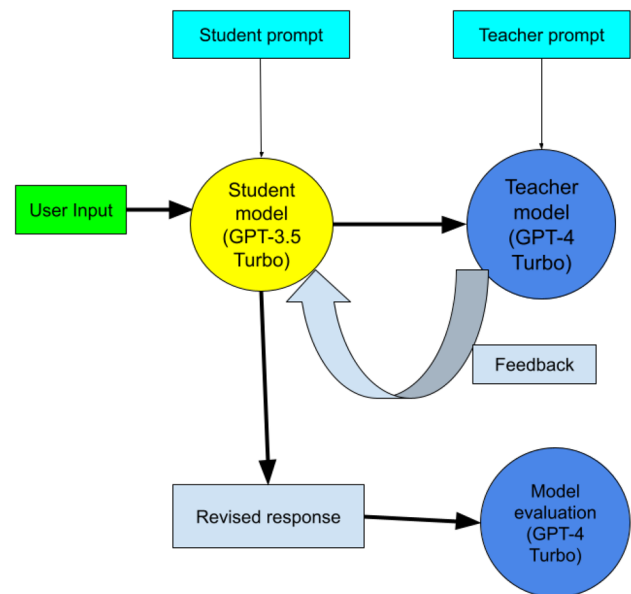
Be warm, empathetic, and constructive

Encourage small, actionable next steps (goal-directed support)

Importantly, do not copy the teacher's wording directly. Instead, write your own improved version that reflects the emotional intelligence shown in the teacher's feedback.

After the student model generated the revised response, the final output was reviewed by GPT-4 Turbo using the EGS evaluation metrics. The framework of the entire process is illustrated in Fig. 1:

The same 273 conversational dialogues were used as input for the self-correction ECoT method. The responses were evaluated by GPT-4 Turbo using the EGS framework, and both the responses and evaluation results were

Fig. 1 Framework of student-teacher model.

stored in a JSON file for subsequent statistical analysis. We also manually annotated 50% of the output data using the EGS metrics ourselves and calculated the kappa agreement score between the human annotations and the model's evaluation results. For the revised ECoT method, the quadratic weighted kappa of 0.77 indicates substantial agreement between human and model ratings, while the self-correction ECoT results show an even higher kappa score of 0.82, reflecting strong alignment.

Component 3: Direct preference optimization (DPO)

The third stage of the EIPO framework is preference optimization: the process by which the student model internalizes the teacher's refinement patterns directly into its parameters. To validate the effectiveness of DPO training combined with self-correction for enhancing the emotional intelligence of LLMs, a final experiment was conducted. We choose Qwen2.5-7B-Instruct as the student model because it can be effectively fine-tuned with both code and data, and our previous research has demonstrated its efficiency as an LLM for procrastination avoidance. We choose GPT-4-turbo as the teacher model because, as shown in the previous section, it has proven to be effective in enhancing the emotional intelligence of LLMs for mitigating procrastination. The student model generates a response to a user's behavior-related message about procrastination, which is then reviewed and refined by the teacher model to produce a more emotionally intelligent version. The prompt for the student model is constructed using the ECoT method from the previous section.

The dataset used, as mentioned above, consists of 273 items focused on procrastination-related conversations. We then combined this dataset with Loughborough University's procrastination questionnaire⁵⁶, which examined the frequency and forms of procrastination, as well as the factors influencing procrastination behavior, using responses collected from 155 staff members and students. And the prompt for the teacher model can be referred to the teacher model prompt in Experiment 2.

The outputs of the student and teacher models are combined to construct a DPO training dataset for fine-tuning the student model. In this dataset, each input conversation is designated as the "prompt", the response generated by the student model is labeled as the "rejected" item, and the refined response produced by the teacher model is labeled as the "chosen" item. This structure allows the DPO framework to leverage pairwise preference signals, encouraging the student model to learn from the teacher's emotionally intelligent refinements and gradually align its responses to the teacher better.

We then applied DPO to fine-tune the Qwen model using the constructed DPO dataset. By training on preference-labeled examples, the model learns to increase the likelihood of preferred outputs, while reducing the probability of less desirable alternatives, thereby improving alignment with the desired conversational style. Due to resource constraints for fine-tuning, we employ the Low-Rank Adaptation (LoRA) method to efficiently fine-tune the Qwen model using the DPO dataset. LoRA keeps the pre-trained model's weights fixed while inserting trainable low-rank matrices into each Transformer layer, significantly lowering the number of parameters that need to be updated for downstream tasks⁵⁷.

In this experiment, LoRA was applied exclusively to the query and value projection submodules of the Transformer self-attention mechanism, as these layers are critical for information flow and contextual representation learning. For DPO training, the model was trained for six complete passes over the training dataset to ensure sufficient exposure to the preference-labeled data. Training metrics were recorded every 10 update steps to monitor convergence and optimization stability. A relatively small constant learning rate was employed to promote stable convergence during preference-based optimization.

The algorithm of the training process is demonstrated in Algorithm 1

```

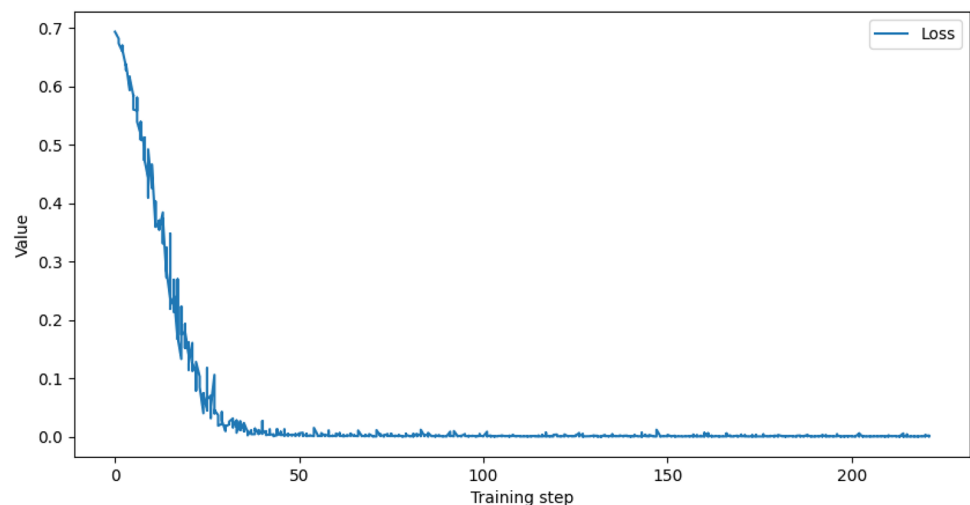
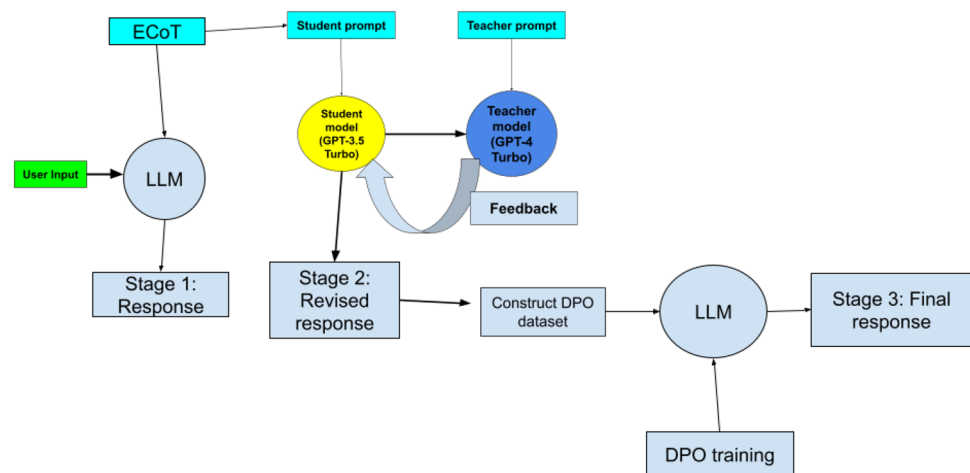
1: Define LoggingDPOTrainer
2:   Extend DPOTrainer
3:   Store training loss and KL divergence history
4:   Override compute_loss:
5:     Compute loss via parent class
6:     Record loss with step index
7:     Compute KL divergence between model logits and reference logits
8:     Record KL divergence
9: Load Pre-trained Model and Tokenizer
10:  Model: Qwen2.5-Coder-7B-Instruct
11:  Set pad token = EOS token
12:  Enable gradient checkpointing
13: Configure LoRA
14:  Set rank  $r = 8$ ,  $\alpha = 16$ , dropout = 0.05
15:  Apply LoRA to query/value projection matrices
16:  Attach LoRA adapters
17: Load Dataset
18:  Load JSON file with preference-labeled DPO data
19: Set Training Arguments
20:  Epochs = 6, batch size = 2, gradient accumulation = 4
21:  Logging interval = 10 steps
22:  Save checkpoints every 200 steps (limit = 2)
23:  Learning rate =  $5 \times 10^{-5}$ , FP16 enabled,  $\beta = 0.1$ 
24: Initialize Trainer
25:  Initialize LoggingDPOTrainer with model, dataset, config
26: Train Model
27:  Compute DPO forward + backward passes
28:  Update only LoRA parameters
29:  Log loss and KL divergence

```

Algorithm 1 DPO training with LoRA on Qwen

Figure 2 presents the training loss trajectory across optimization steps. The model demonstrates a rapid reduction in loss during the initial phase, with values declining from approximately 0.7 to below 0.1 within the first 40 steps. This steep decrease indicates that the optimization procedure effectively captured informative gradients early in training. Afterwards, the loss continued to decrease more gradually and remaining stable thereafter. And such pattern reflects both efficient convergence and stability of the learning process.

After obtaining the fine-tuned model, we used combined conversational dialogues as input to evaluate the outputs of both the baseline Qwen-2.5-7B and the fine-tuned Qwen-2.5-7B models. The responses were assessed by GPT-4 Turbo using the EGS framework, and both the generated responses and the evaluation results were stored

Fig. 2 Training loss over DPO optimization steps.**Fig. 3** Overview of the proposed emotionally intelligent preference optimization (EIPO) framework.

in a JSON file for subsequent statistical analysis. Subsequently, a win rate analysis was conducted to quantify the comparative performance of the fine-tuned model against the baseline Qwen-2.5-7B. The win rate was measured using GPT-4 Turbo as the evaluation model. The win rate (WR) can be defined as:

$$\text{Win Rate (\%)} = \frac{N_{\text{win}}}{N_{\text{total}}} \times 100$$

Where:

N_{win} = number of comparisons where the model's output is judged better by GPT-4 Turbo

N_{total} = total number of pairwise comparisons evaluated

Figure 3 illustrates the overall pipeline of the proposed EIPO framework.

Below is an example of response progression across EIPO stages:

User: "I have a test coming up and I should be studying right now, but I procrastinate studying."

System response from Stage 1 – Emotional Reasoning (ECOT) "It's totally understandable to feel that way! Procrastination can be tough, but you've got this! Maybe try setting small study goals or creating a study schedule to make it more manageable. You're capable of doing well on the test, so don't be too hard on yourself. Good luck with your studying!"

System response from Stage 2 – self-correction (teacher refinement) “It sounds like you’re in a bit of a relatable bind with your test! Procrastination can definitely sneak up on us. Maybe breaking your study time into smaller, more manageable chunks could help ease the pressure? And don’t forget, it’s perfectly okay to give yourself little breaks to recharge, maybe even chuckle over our chat here! What’s the subject of the test? Maybe I can help with some quick tips or resources!” System response from Stage 3 – preference optimization (DPO-finetuned model) “I understand that facing a big test can feel overwhelming, but remember, taking small steps towards preparation can make a huge difference! Maybe try setting aside just a little bit of time each day to review material instead of cramming everything at once. This approach not only helps with retention but also reduces last-minute stress. Plus, breaking tasks into manageable pieces makes them less daunting. How does that sound?”

Cross-model generalization and independent model evaluation

While the proposed EIPO framework was developed and evaluated using GPT-3.5 Turbo, Qwen2.5-7B-Instruct, and GPT-4 Turbo, we further validate its model-agnostic design through cross-model generalization experiments. This addresses two critical concerns: (1) whether EIPO’s components—ECoT prompting, teacher-guided self-correction, and DPO fine-tuning—transfer effectively to alternative LLM architectures, and (2) whether evaluation results are robust to the choice of the evaluator model.

A key advantage of the proposed EIPO framework is its architecture-agnostic nature. While our primary experiments utilized Qwen2.5-7B-Instruct, we further demonstrate generalizability by applying the ECoT prompting and DPO fine-tuning pipeline to two additional, widely-used open-weight models: Llama-3.2-3B-Instruct and Gemma-2-2B-IT.

Following the identical protocol from Section [Component 3: Direct preference optimization \(DPO\)](#), we applied the ECoT prompt to both models to generate baseline responses on the procrastination dialogue dataset. Subsequently, we fine-tuned each model using LoRA and the DPO dataset constructed from teacher-model (GPT-4 Turbo) refinements. Performance was evaluated using the EGS framework with GPT-4 Turbo.

To address the concern that using GPT-4 Turbo for both generating feedback (as the teacher) and evaluating performance (EGS scoring) may introduce circular reasoning, we implemented a complementary validation strategy.

To provide a completely independent automated evaluation, we replaced GPT-4 Turbo with Claude-3.5-Sonnet to re-evaluate all responses from the primary Qwen experiments, using the identical EGS prompt. Claude-3.5-Sonnet was selected because it is architecturally distinct from OpenAI’s models, reducing the risk of systematic bias.

Results

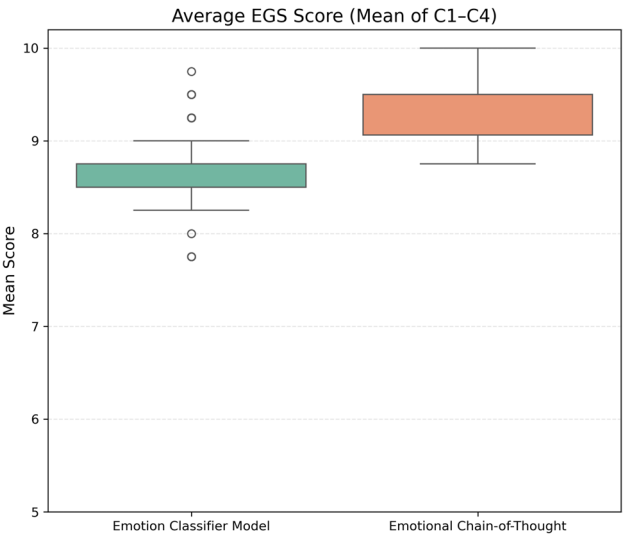
We evaluate the EIPO framework through a component-wise ablation design. Each successive research question isolates the marginal contribution of one framework stage: emotional reasoning (RQ1), self-correction (RQ2), and preference optimization (RQ3). All evaluations use the emotional generation score (EGS) framework with GPT-4 Turbo as the automated evaluator, unless otherwise noted. Each EGS criterion—context relevance (C1), emotional safety (C2), positive emotion (C3), and constructive contribution (C4)—is scored 0–10.

To establish the reliability of automated evaluation, we conducted human validation on a randomly sampled 20% subset of model outputs. Quadratic weighted Cohen’s κ showed substantial agreement between human and model ratings for both the classifier-conditioned baseline ($\kappa = 0.7368$) and ECoT ($\kappa = 0.7589$). These scores confirm that GPT-4 Turbo’s EGS evaluations are a trustworthy proxy for human judgment, enabling consistent, scalable assessment across all comparisons.

Table 1 Mean EGS scores (standard deviations in parentheses) for the emotion classifier model and emotional chain-of-thought.

Compared	Context	Emotional	Positive	Constructive
Method	Relevance	Safety	Emotion	Contribution
Emotion classifier model	8.45 (0.60)	9.62 (0.51)	8.47 (0.71)	7.96 (0.46)
Emotional chain-of-thought	9.72 (0.46)	9.99 (0.06)	9.09 (0.41)	8.78 (0.44)

Fig. 4 Average emotional grounding scores (EGS) by method. The emotional chain-of-thought (ECoT) yields higher median scores and more consistent performance than the emotion classifier model, indicating improved emotional understanding and generation.



RQ1: Does emotional reasoning improve response quality?

All 273 conversational dialogues were used as input for each method. The corresponding C1–C4 scores from model evaluation were extracted from the results, and the average score for each criterion is presented in the table below.

Table 1 reports mean EGS scores. ECoT outperforms the classifier-conditioned baseline on every criterion. The largest gains are observed in constructive contribution ($\Delta = +0.82$) and Context Relevance ($\Delta = +1.27$), with the latter representing a particularly substantial improvement. Positive emotion also demonstrates a meaningful increase ($\Delta = +0.62$). A paired *t*-test confirms that all improvements are statistically significant ($p < 0.0001$). Furthermore, ECoT consistently exhibits lower standard deviations across all evaluation criteria, indicating more stable and consistent performance than the classifier-conditioned baseline.

Figure 4 displays the distribution of average EGS scores. ECoT achieves a higher median than the classifier baseline. The minimum ECoT score (8.7) exceeds the classifier baseline’s first quartile, demonstrating that structured reasoning improves not only average performance but also consistency.

Constructive Contribution remains the lowest-scoring criterion for both methods—a persistent challenge for empathetic AI. We iteratively refined the ECoT prompt’s Step 5 (Influencing Others’ Emotions) to emphasize goal-directed facilitation alongside emotional validation. The revised prompt tasks the model with “reducing distress, increasing motivation, or building connection.” Re-evaluation on the same 273 dialogues yields a mean C4 score of 9.02, an improvement of +0.24 over the original ECoT. Prompt engineering within the emotional reasoning stage can directly strengthen specific EGS dimensions.

RQ2: Does self-correction further enhance response quality?

We compared the average EGS scores between the revised emotional chain-of-thought (ECoT) approach presented in the previous section and the Self-correction Emotional Chain-of-Thought (Self-correction ECoT)

Fig. 5 Comparison of average EGS scores by ECoT and self-correction ECoT.

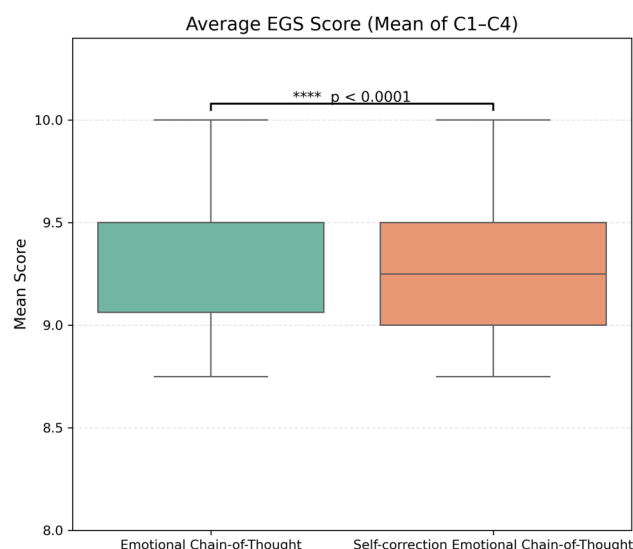


Table 2 Comparison of mean EGS scores between the baseline and DPO-fine-tuned Qwen models. Values are reported as mean (standard deviation).

Compared	Context	Emotional	Positive	Constructive
Method	Relevance	Safety	Emotion	Contribution
Baseline Qwen (ECoT)	9.23 (0.50)	9.94 (0.23)	8.92 (0.65)	8.54 (0.69)
DPO-fine-tuned Qwen	9.43 (1.02)	10.00 (0.00)	8.96 (0.71)	8.76 (0.95)

Significant values are in bold.

approach. A paired t -test was conducted on the mean EGS scores, yielding a statistically significant difference between the two methods ($p < 0.0001$). Figure 5 presents the boxplot comparing the distribution of average EGS scores for the two approaches.

The self-correction emotional chain-of-thought (SC-ECoT) method achieved a marginally higher mean EGS score (9.50) compared to the base emotional chain-of-thought (ECoT) approach (9.45), representing a modest improvement of +0.05 points. The modest improvement from self-correction (+0.05) suggests diminishing returns when the base model already performs at a near-ceiling level, indicating that self-correction may be more beneficial for challenging samples or lower-performing models.

Overall, SC-ECoT demonstrates a slight but consistent improvement over ECoT, achieving a mean score of 9.50 compared to 9.45. These findings suggest that self-correction can provide incremental benefits, though the magnitude of improvement is limited by the already high performance of the base method.

RQ3: Does preference optimization stabilize and improve gains?

Table 2 reports mean EGS scores. The DPO-fine-tuned model outperforms the baseline on all four criteria. The largest gain is again in constructive contribution ($\Delta = +0.22$), confirming that preference optimization is particularly effective at embedding the teacher's capacity for action-oriented support.

Critically, DPO stabilizes the gains introduced by self-correction. We created the boxplot for results, and the boxplot illustrating the average EGS scores for the Qwen-2.5-7B and fine-tuned model is shown in Fig. 6.

From the median performance, the DPO fine-tuned model performs better according to the EGS metric. The DPO fine-tuned model shifts the central tendency upward and achieves higher top-end scores. Overall, the DPO fine-tuning appears effective in improving emotional intelligence in procrastination avoidance scenario as measured by EGS, raising both the median and maximum scores. However, the slightly wider spread suggests that further refinement might be needed to stabilize performance to produce optimal responses.

Fig. 6 Comparison of average EGS scores by Qwen-2.5-7B and DPO.

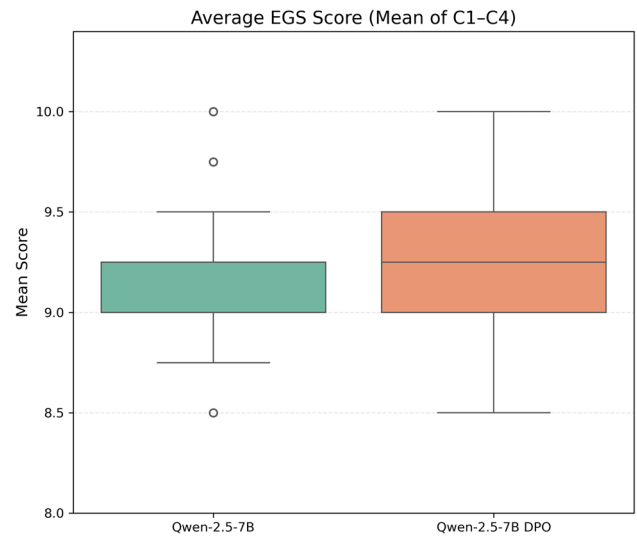


Table 3 Cross-model generalization: mean $\pm$ SD EGS scores before and after EIPO fine-tuning.

Student model	Baseline (ECoT)	DPO-finetuned	Δ (EGS)
Qwen2.5-7B-Instruct	9.16 $\pm$ 0.52	9.29 $\pm$ 0.67	+0.13
Llama-3.2-3B-Instruct	8.92 $\pm$ 0.58	9.18 $\pm$ 0.62	+0.26
Gemma-2-2B-IT	8.85 $\pm$ 0.61	9.12 $\pm$ 0.59	+0.27

We also calculated the win rate of the fine-tuned model using the procrastination-related dialogue dataset, allowing direct comparison with the baseline model on task-specific conversational abilities. A total of 428 pairwise comparisons were conducted to assess the relative quality of the two responses from baseline model and fine-tuned model. The analysis revealed that responses from baseline model achieved a win rate of 36.82%, while response from fine-tuned model achieved a higher win rate of 63.18%. This outcome indicates that response from DPO model was favored more often than response from baseline model across the EGS evaluation metrics. Overall, response from DPO fine-tuned model demonstrates a performance advantage in terms of emotional intelligence within the context of procrastination avoidance scenario.

Representative examples of responses generated by the standard Qwen2.5-7B-Instruct model and the DPO-fine-tuned model are presented below.

User: “Man, don’t you hate when you have a test coming up and you procrastinate studying?”

Response from standard Qwen2.5-7B-Instruct model: “Hey there! It sounds like you’re feeling a bit overwhelmed with your upcoming test. Procrastination can definitely make things feel even more stressful. Remember, taking small steps towards studying every day can make a big difference in how prepared you feel. If you ever want to talk about strategies or just need someone to vent to, I’m here for you. Let’s make sure you’re ready and relaxed come test day!”

Response from DPO-fine-tuned model: “I completely understand how frustrating it can be to feel like you’re running out of time to prepare for a test. It’s totally normal to feel overwhelmed sometimes, especially when deadlines loom large. Remember, taking small, manageable steps each day can make a big difference in how prepared you’ll feel come test day. Maybe try setting aside just a little bit of time each hour to review your notes? That way, you won’t feel so rushed, and you’ll gradually get closer to being fully prepared. If you ever want to talk about strategies or just vent, I’m here to listen. Good luck with your studies!”

Findings for cross-model generalization and independent model evaluation

Table 3 reports the Mean $\pm$ SD EGS scores. The results show that both alternative models achieve significant performance improvements after DPO fine-tuning, confirming that EIPO's staged optimization—emotional reasoning, self-correction, and preference learning is not specific to a single model family. Notably, smaller models like Gemma-2-2B demonstrate larger relative gains ($\Delta = +0.27$), suggesting that the framework is particularly effective at enhancing the emotional intelligence of more compact LLMs, which are often preferred for deployment in resource-constrained CUIs due to lower latency and computational cost.

A paired *t*-test confirmed that all improvements were statistically significant ($p < 0.001$). These results empirically validate the claim that EIPO's components are modular and transferable across model families, reinforcing the framework's practical utility for deployment in diverse conversational AI systems.

For independent LLM evaluator, the agreement between the two LLM evaluators is exceptionally high (Pearson's $r = 0.91$). We further computed the win-rate comparison between baseline Qwen and DPO-fine-tuned Qwen using Claude-3.5-Sonnet as the evaluator. The fine-tuned model achieved a win rate of 62.1%, closely matching the 63.18% win rate obtained with GPT-4 Turbo (Section RQ3: Does preference optimization stabilize and improve gains?). This consistency across independent evaluators provides strong evidence that the observed improvements are genuine and not an artifact of a single model's evaluation bias.

Discussion

The results demonstrate that the EIPO framework systematically improves the emotional intelligence and supportiveness of conversational responses through a staged combination of emotional reasoning, self-correction, and preference optimization. Each component contributes distinct and complementary benefits, highlighting the importance of integrating reasoning and alignment mechanisms when designing emotionally intelligent conversational agents for procrastination avoidance.

First, emotional reasoning emerges as the primary driver of improvement in emotional response quality. The emotional chain-of-thought (ECoT) method significantly outperforms the classifier-conditioned baseline across all EGS criteria, with the largest gains observed in positive emotion and constructive contribution. These improvements are particularly important in procrastination-related contexts, where effective support must both validate emotional experiences and encourage actionable progress. The improved consistency and higher minimum scores further suggest that structured emotional reasoning enables more reliable and contextually appropriate responses.

Second, self-correction introduces an effective mechanism for refining emotional responses and generating high-quality alignment signals. While self-correction increases the maximum achievable performance, it also introduces variability due to occasional overcorrection or misinterpretation of feedback. This suggests that iterative refinement alone may not guarantee stable improvements during inference. However, the key contribution of self-correction lies in its ability to generate preference-aligned response pairs, which serve as valuable training data for preference optimization. This highlights the role of self-correction as an intermediate alignment mechanism that enables scalable and structured emotional improvement.

Third, preference optimization using DPO consolidates and stabilizes gains achieved in earlier stages. The DPO-fine-tuned model consistently outperforms the baseline across all emotional quality dimensions, with particularly strong improvements in constructive contribution. This suggests that preference learning effectively integrates action-oriented emotional support into the model's response generation capabilities. The upward shift in median and maximum performance, along with the higher win rate in pairwise comparisons, confirms that preference optimization produces more consistently preferred responses. Although minor performance variability remains, the overall improvements demonstrate that preference optimization successfully transforms iterative emotional refinement into durable and generalizable capability improvements.

Taken together, these findings validate the staged design of the proposed emotionally intelligent preference optimization (EIPO) framework. Rather than functioning as independent components, each stage contributes a complementary role within a progressive optimization process. Emotional chain-of-thought establishes explicit emotional reasoning to generate contextually appropriate and supportive responses, teacher-guided self-correction converts this reasoning into high-quality preference supervision, and Direct Preference Optimization internalizes these learned preferences into the model parameters, resulting in more robust and consistent emotional behavior during inference. This staged optimization systematically transfers emotional reasoning from prompt-level guidance to intrinsic model capability, leading to cumulative improvements in emotional expressiveness, empathy, and constructive guidance. Consequently, EIPO provides a principled methodology for developing emotionally intelligent conversational agents that extends beyond conventional prompt engineering-based approaches.

The cross-model generalization experiments further validate EIPO's model-agnostic design. The consistent improvements observed on Llama-3.2-3B and Gemma-2-2B, with particularly pronounced gains on smaller models, suggest that EIPO's staged optimization can effectively enhance emotional intelligence across diverse architectures. Furthermore, the high agreement between GPT-4 Turbo and Claude-3.5-Sonnet evaluations mitigates concerns about circular reasoning and supports the reliability of automated EGS assessment as a scalable evaluation methodology.

More broadly, the results underscore the importance of integrating reasoning, feedback, and preference learning in the design of CUIs to foster emotionally intelligent behavior-change support. In procrastination-related interactions, effective CUIs must provide not only emotional validation but also actionable guidance that encourages motivation and sustained task engagement. The EIPO framework offers a principled design approach for CUIs, demonstrating how emotionally informed reasoning and preference optimization can enhance both the emotional quality of user interaction and the practical effectiveness of conversational systems in supporting behavior change.

Conclusion and future work

This paper introduced emotionally intelligent preference optimization (EIPO), a staged framework for enhancing the emotional intelligence of LLMs in procrastination-related conversations. Through component-wise ablation, we showed that emotional reasoning delivers the largest gains in emotional support and constructive guidance, self-correction generates high-quality preference signals, and direct preference optimization (DPO) consolidates these gains into more stable and consistently preferred responses.

Together, these findings demonstrate that integrating structured emotional reasoning with preference learning offers a principled approach to enhancing emotional intelligence in CUIs designed to support procrastination avoidance. The consistent performance gains observed across Qwen2.5-7B, Llama-3.2-3B, and Gemma-2-2B confirm that EIPO's core components, ECoT prompting, teacher-guided self-correction, and DPO, transfer effectively across diverse architectures, making the framework suitable for deployment in a wide range of conversational systems.

Future research could expand on these findings in several directions. Firstly, integrating multimodal emotional cues such as voice tone, facial expressions, or typing behavior may enhance the model's capacity to detect and respond to users' emotional states more accurately. Secondly, we can investigate the causes of the low-scoring outliers in ECoT responses, aiming to identify conditions under which the method underperforms and to develop strategies to improve consistency across all outputs. Besides, future work will investigate the generalizability of the proposed EIPO framework through cross-domain and zero-shot evaluations on other behavior-change applications, such as stress management, anxiety support, and healthy habit formation. Finally, the future longitudinal user study is vital to evaluate how emotionally intelligent chatbots influence real-world behavior change and motivation over time.

Author contributions B.C. wrote the main manuscript text. N.D. reviewed the manuscript. All authors reviewed the final version of the manuscript.

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Data availability The EmpatheticDialogues dataset used in this study is publicly available: https://huggingface.co/datasets/facebook/empathetic_dialogues. The dataset contains 24,850 conversations centered on emotionally grounded situation descriptions collected from 810 participants.

Code availability The source code supporting this study is publicly available at GitHub and archived on <https://doi.org/10.5281/zenodo.22212997>

Declarations

Competing interests The authors declare no competing interests. "<https://doi.org/10.5281/zenodo.22212997>"

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